Investigating thematic role salience with a non-linguistic change-detection task

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Thematic roles (e.g. agent, patient, source, goal) are foundational constructs in our understanding of the syntax-semantics interface and language acquisition.[1,2] Certain roles are more linguistically prominent than others, either in terms of their production probability (some are mentioned more frequently[3]) or in terms of structural dominance (some occupy higher syntactic positions[4]). Prior work using a change-detection paradigm found that people were more likely to miss changes involving the source of a motion event than the goal, mirroring the difference in production probability.[5-7] Such findings have led theorists to suggest that linguistic role asymmetries could arise from differences in the perceptual salience or memorability of event participants.[5,6,8] Building on this approach, we examined whether parallel memory effects are present for other role pairs that differ in either production probability (goal > source; theme > instrument) or structural dominance (agent > patient; donor > recipient).

Across two different experiments, English speaking adults (exp 1: n = 62, exp 2, online replication: n= 187) completed a change-detection task modeled after Papafragou's task[5]. There were four conditions that differed in terms of event type and the roles that were contrasted: source-goal, theme-instrument, agent-patient, and donor-recipient events (Fig-1). Each participant completed either four conditions (exp 1) or two conditions (exp 2), blocked, with order counterbalanced. For each condition, participants first had viewed a series of short videos depicting events of that type (e.g., a baby crawling from a desk to a bed), followed by the test phase consisting of four trial types: no-change controls, noticeable background changes, changes to the prominent role, and changes to the less prominent role. The critical role changes involved different objects from the same category (e.g. a source change would involve a desk changing to a different desk, see Fig-1). Objects were rotated across lists so that the changes for prominent and non-prominent roles were identical. Four training trials preceded each condition to familiarize participants with the task. In both experiments, across all conditions, participants performed well on control items, noticing the background changes and correctly accepting the no-change trials as familiar (see Fig-2). On the critical role-change trials, performance was close to ceiling in exp-1, and neither at ceiling nor floor in exp-2, but the within-condition comparisons revealed no significant differences between prominent and less prominent roles (for both exp-1 and exp-2, $\beta s<.7$, $ps>.1$, see Fig-2).

Thus we find no evidence for differences in change detection based on role prominence. Surprisingly, we even failed to replicate the source–goal asymmetry. To enable a more direct comparison with previous studies,[5,8] we conducted exp 3 (n=30, English speaking adults, sample size determined via an a priori power analysis, $\alpha = .05$, power = .80, based on the effect size reported by Papafragou[5], d = 1.08) using only the source–goal trials and omitted the training trials to match earlier procedures. In addition, we reanalyzed the source–goal trials from the first block of exp 1 and 2. Across these three datasets (exp 3 and the reanalyses of exp 1 and 2), we found no reliable differences between roles ($\beta s < .4$, $ps > .2$, see Fig-3). The only apparent difference between this task and Papafragou[5]'s is that ours was conducted online. We interpret these results in light of the broader literature in which the effects of role prominence on memory vary with task, event type, and thematic role.[6-8] Taken together, this pattern suggests that the asymmetries observed in language are unlikely to arise from differences in the perceptual salience or memorability of event roles. While these findings are consistent with both theories that treat thematic roles as purely linguistic constructs, they are also consistent with theories that attribute them –and the linguistic asymmetries– to underlying cognitive representations. Indeed, there is recent evidence that thematic role concepts are available to people with no access to an existing language.[9] Production probability may reflect the importance of communicating about a role, while syntactic dominance could reflect hierarchical event structure[4] or the argument's position in a causal chain.[10]

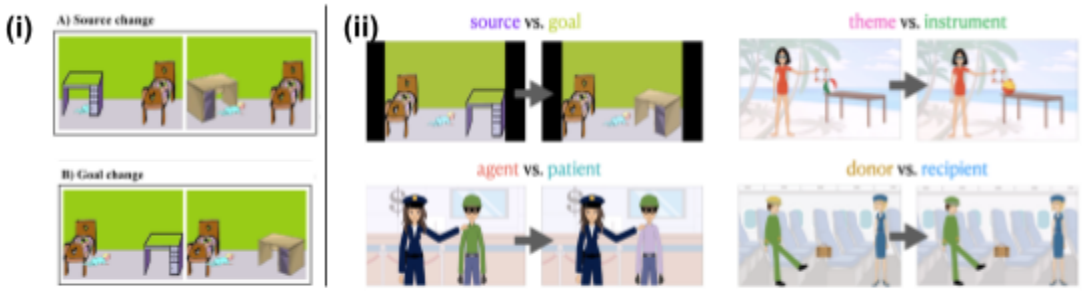

Figure 1: (i) Example of a source change (a) and a goal change (b). Stimuli for the source-goal condition were adapted from Papafragou (2010) **(ii)** Example of a change in source-goal condition, theme-instrument, agent-patient condition, and donor-recipient condition.

Figure 2: Proportion of detected changes in study 1 (upper panel) and study 2 (lower panel) for the **(i)** Source-Goal condition, **(ii)** Theme–Instrument condition, **(iii)** Agent–Patient condition, and **(iv)** Donor–Recipient condition. There were four types of trials: two control trials (change in the background, the red boxplot; and no change, the green boxplot), a change in the more prominent role (blue) and a change in the less prominent role (purple). The black line represents the median, and the red line, the mean of each change type. Triangles represent individual participants.

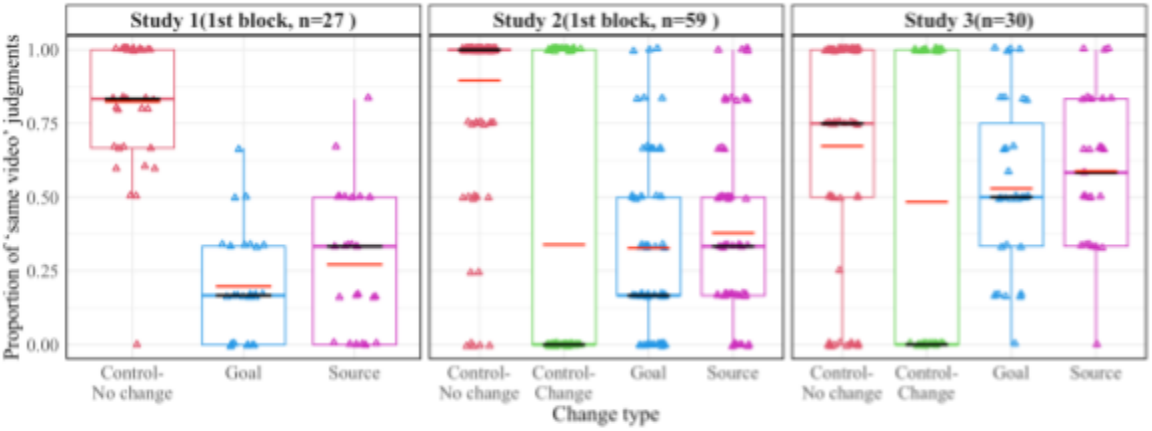

Figure 3: Proportion of detected changes in the Source-Goal condition in the first block of study 1 (upper panel) and study 2 (middle panel) and in study 3 (lower panel). The black line represents the median, and the red line, the mean of each change type. Triangles represent individual participants.

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